



DeepLearning.AI

Data Modeling, Transformation and Serving

Serving Data



DeepLearning.AI

Serving Data for Analytics and Machine Learning

Week 4 Overview

Serving Data — Analytics Use Cases

Business Intelligence



Dashboards

Reports



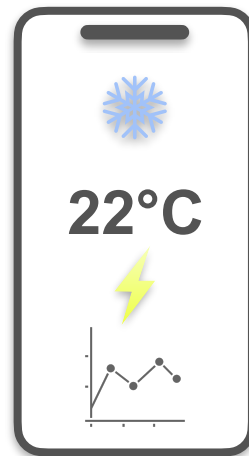
Operational Analytics

Monitor data to inform
immediate action



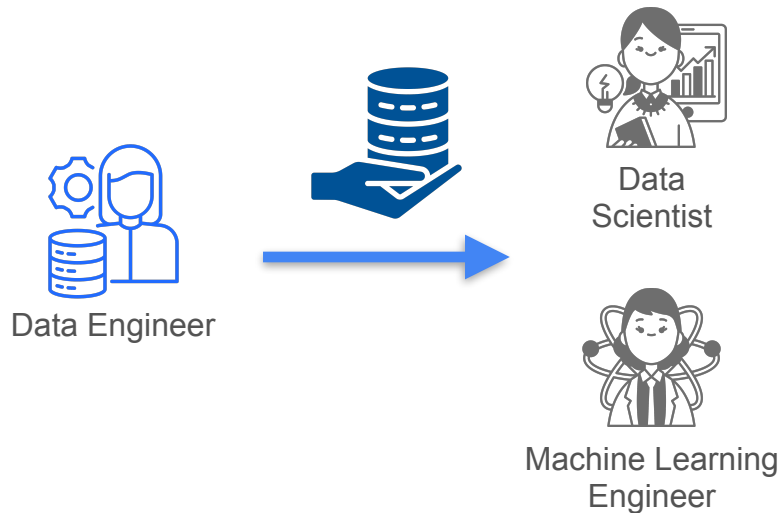
Serve data within the
required latency

Embedded Analytics



Serving Data — Machine Learning Use Cases

Becoming widely adopted in many companies



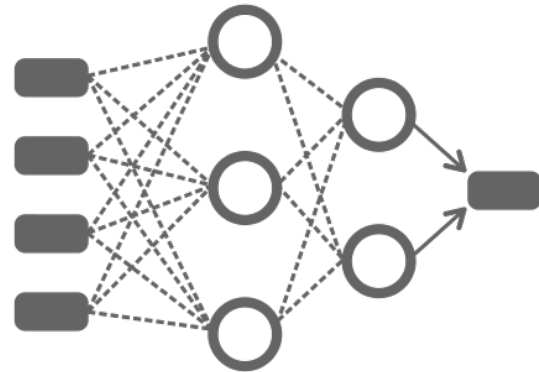
Model training

FEATURE 1

FEATURE 2

FEATURE 3

FEATURE 4



Example:

- Customer churn model
- Recommendation system

Serving Data



Stakeholders



Semantic Layer

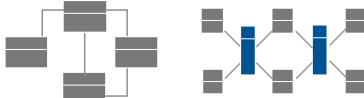
- Document definitions (e.g. meaning of an active user)
- Derive business metrics (e.g. revenue)
- Create a common language for data



Data Sources



Data Model



You can serve data as...

A table

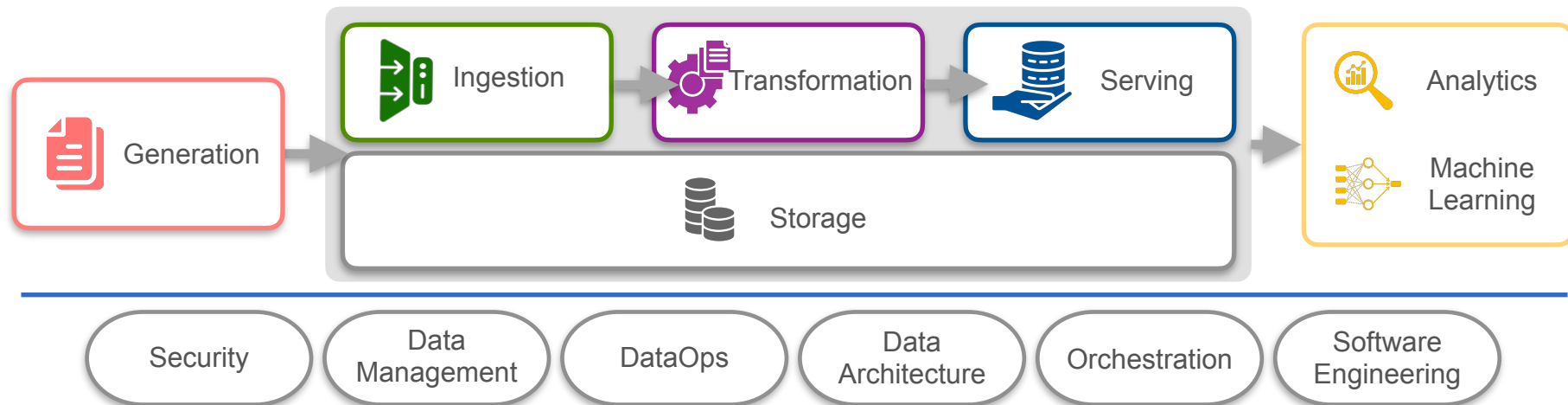
View

Materialized
View

Thinking Like a Data Engineer



Data Engineering Lifecycle and Undercurrent



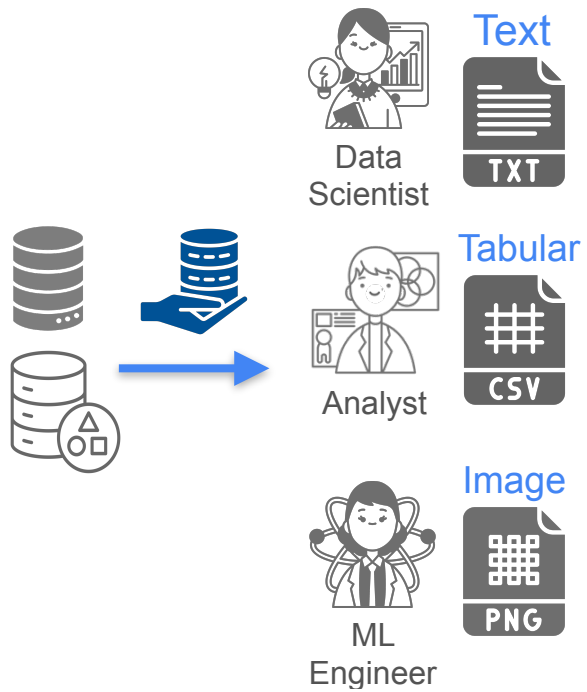


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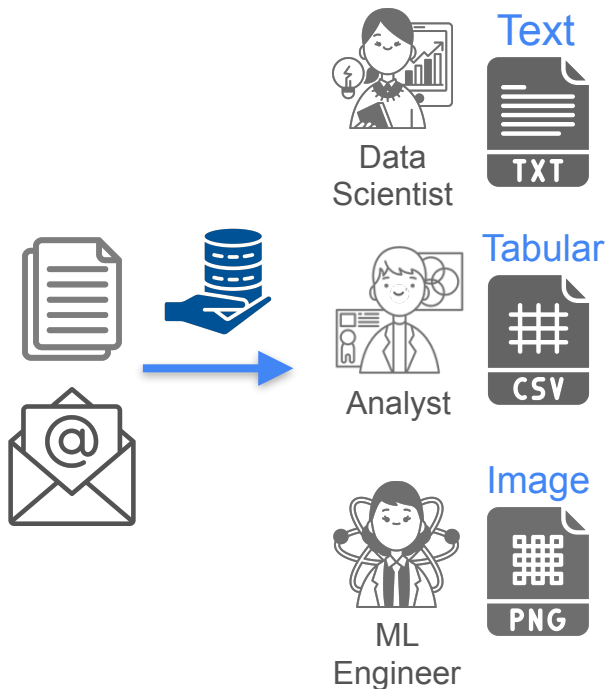
Serving Data for Analytics and Machine Learning

Serving Data

As Files



As Files



As Files

For ad hoc requests



Data
Scientist

Text



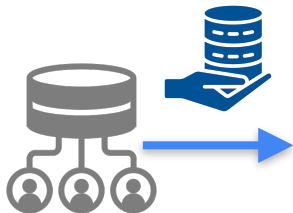
Analyst

Tabular



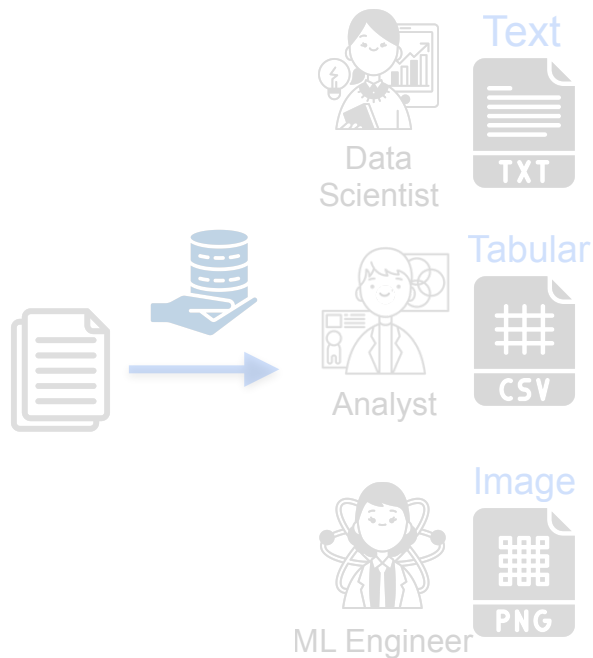
ML Engineer

Image

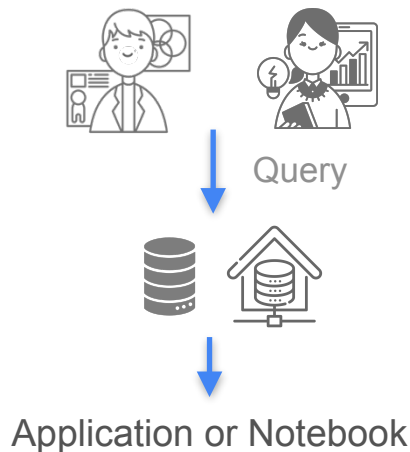


As Files

For ad hoc requests



From Databases and Data Warehouses

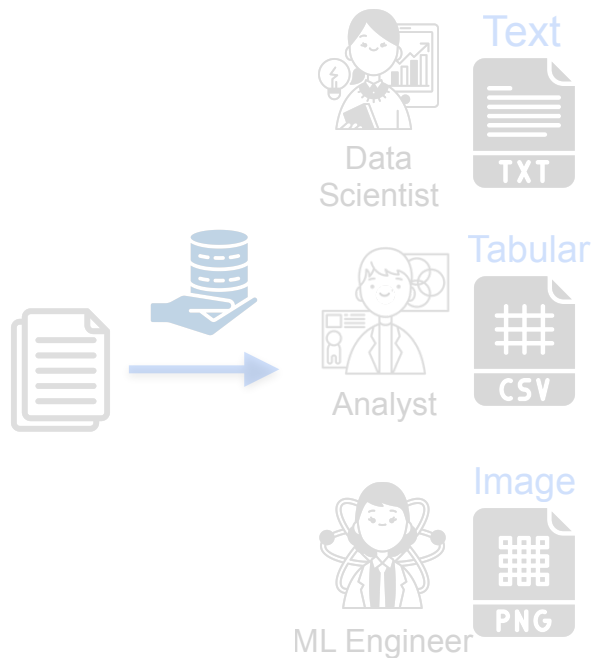


Benefits:

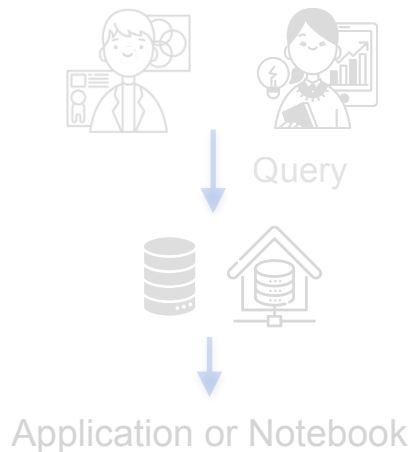
- Imposes order & structure through schema
- Gives you fine-grained permissions controls
- Offers high performance for queries

As Files

For ad hoc requests



From Databases and Data Warehouses

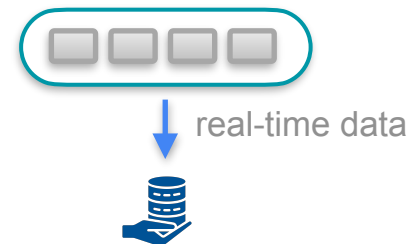


Benefits:

- Imposes order & structure through a schema
- Gives you fine-grained permissions controls
- Offers high performance for queries

From Streaming Systems

Streaming Systems



Example

 Operational analytics database

- Enables low latency analytical queries across historical and current data
- Effectively combining the features of an OLAP database with a stream-processing system

Data Management



Data Engineer



Trust the data

Can interpret the data correctly



Stakeholders

Ensuring data
correctness,
consistency,
trustworthiness

Data Definition

Meaning of data as it is understood throughout the organization
e.g. document and make available the definition of “Customer”

Data Logic

Consists of formulas for deriving metrics from data
e.g. gross sales or customer lifetime value

Customer
Churn Metric



Understand
the definition
of “Customer”



Write SQL
query



Define this metric so
it can be reused

Semantic Layer

- Translate the underlying data elements and structures into more intuitive business terms
- Ensures consistent definitions for business terms
- Helps end users more easily navigate the data

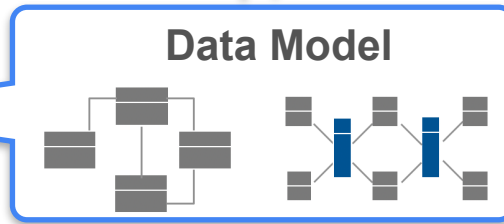
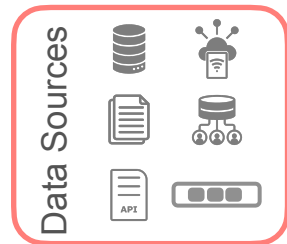
Created in a BI tool

OR

Created using software



(using YAML files & SQL queries)





Stakeholders



Use BI platform to visualize and analyze data



Amazon QuickSight



Use a notebook to explore data, engineer features, or train a model



Amazon SageMaker



Google Cloud Vertex AI



Azure Machine Learning

Semantic Layer

- Translate the underlying data elements and structures into more intuitive business terms
- Ensures consistent definitions for business terms
- Helps end users more easily navigate the data

Created in a BI tool

OR

Created using software



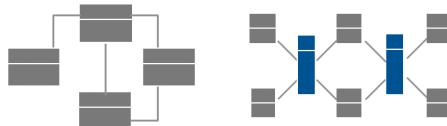
(using YAML files & SQL queries)



Data Sources



Data Model





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Serving Data for Analytics and Machine Learning

Views and Materialized Views

View

- Stored in the database to provide easier access to common queries
- Represents a virtual table, not a physical one
- When selecting from a view:
 - The database creates a new query
 - The query optimizer optimizes and runs the query

```
CREATE VIEW customer_info AS
SELECT
    first_name,
    last_name,
    email,
    phone,
    city,
    postal_code,
    country
FROM
    customer
JOIN address ON address.id = customer.address_id
JOIN city ON city.id = address.city_id
JOIN country ON country.id = address.country_id
```

View

- Stored in the database to provide easier access to common queries
- Represents a virtual table, not a physical one
- When selecting from a view:
 - The database creates a new query
 - The query optimizer optimizes and runs the query

```
CREATE VIEW customer_info AS  
SELECT
```

```
first_name,  
last_name,  
email,  
Phone,  
city,  
postal_code,  
country
```

Effectively restricting data access to
only the data needed



```
FROM
```

```
customer  
JOIN address ON address.id = customer.address_id  
JOIN city ON city.id = address.city_id  
JOIN country ON country.id = address.country_id
```



Marketing Analyst

query



Wide Table:
customer, address, city, country

View

- Stored in the database to provide easier access to common queries
- Represents a virtual table, not a physical one
- When selecting from a view:
 - The database creates a new query
 - The query optimizer optimizes and runs the query
- Database object
- Stored and persists on disk

Common Table Expressions (CTE)

- Represents temporary results that you can reference
- Only exists within scope of the main query where they are referenced

```
WITH name of the CTE AS (  
Query),  
  
Subsequent query
```

View

- Stored in the database to provide easier access to common queries
- Represents a virtual table, not a physical one
- When selecting from a view:
 - The database creates a new query
 - The query optimizer optimizes and runs the query
- Database object
- Stored and persists on disk



Can be expensive
for frequently-run
complex queries

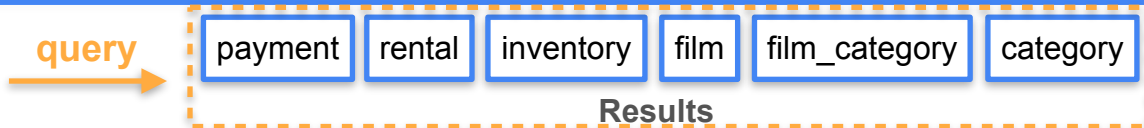
Common Table Expressions (CTE)

- Represents temporary results that you can reference
- Only exists within scope of the main query where they are referenced

Materialized Views

- Does some or all of the view computations in advance
- Caches the query results and allows you to refresh the data
- Some latency between refreshes

```
CREATE MATERIALIZED VIEW rental_by_category AS  
SELECT category.name AS category,  
       sum(payment.amount) AS total_sales  
FROM payment  
      JOIN ... rental, inventory, film, film_category, category ...  
GROUP BY category.name
```





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Data Engineering

Summary of the Program Concepts

Thinking Like a Data Engineer



1

Identify business goals & stakeholder needs

1. Identify business goals & stakeholders you will serve
2. Explore existing systems and stakeholder needs
3. Ask stakeholders what actions they will take with the data product



2

Define system requirements

1. Translate stakeholder needs to functional requirements
2. Define non-functional requirements
3. Document and confirm requirements with stakeholders



3

Choose tools & technologies

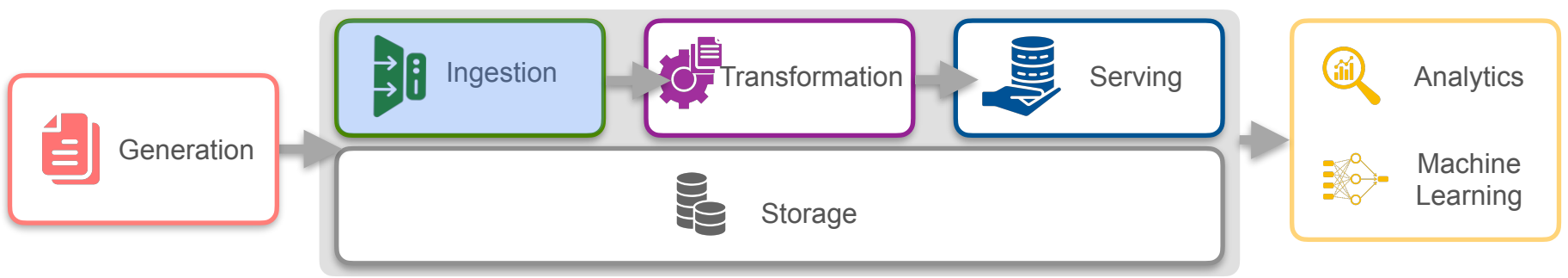
1. Identify tools & tech to meet non-functional requirements
2. Perform cost / benefit analysis and choose between comparable tools & tech
3. Prototype and test your system, align with stakeholder needs

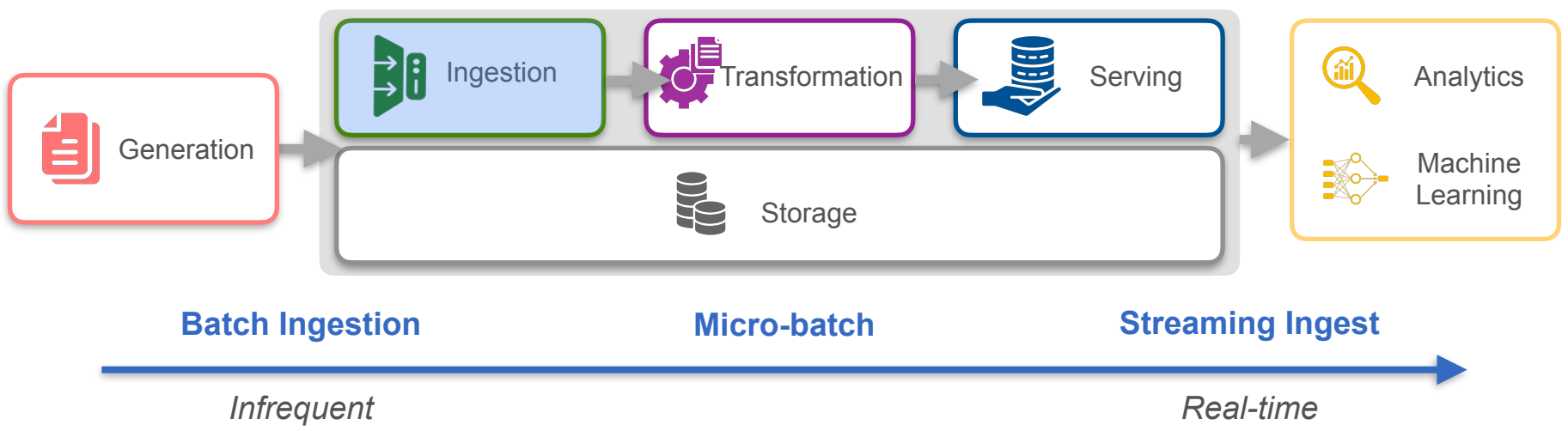


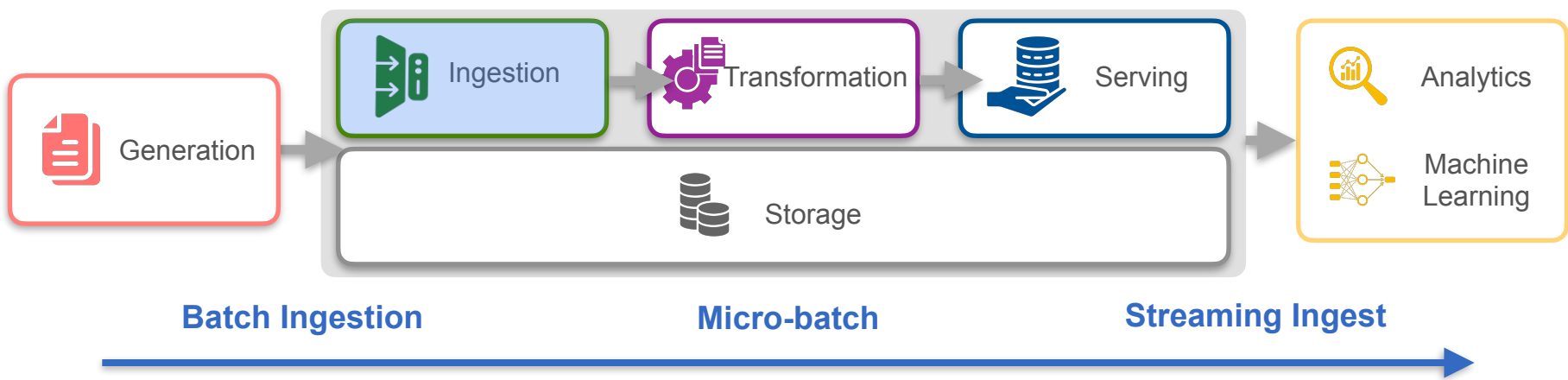
4

Build, evaluate, iterate & evolve

1. Build & deploy your production data system
2. Monitor, evaluate, and iterate on your system to improve it
3. Evolve your system based on stakeholder needs



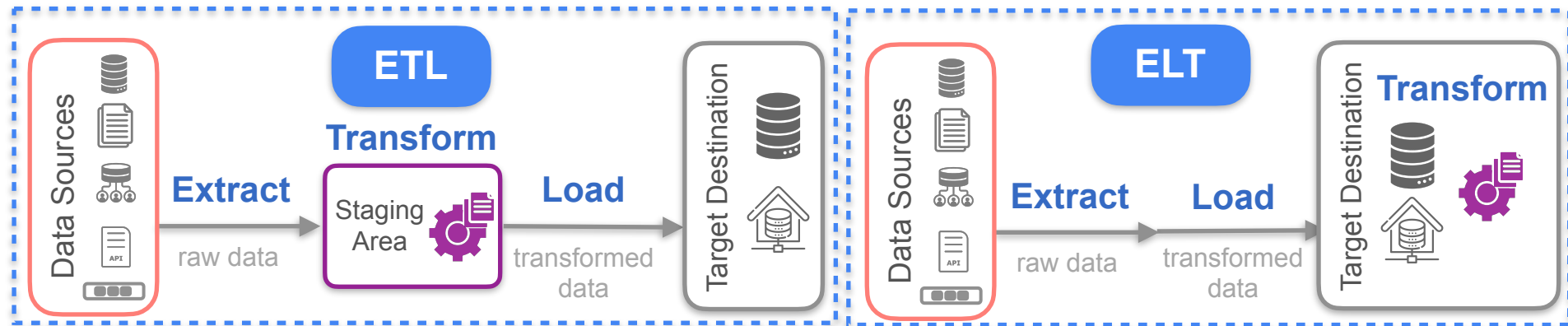


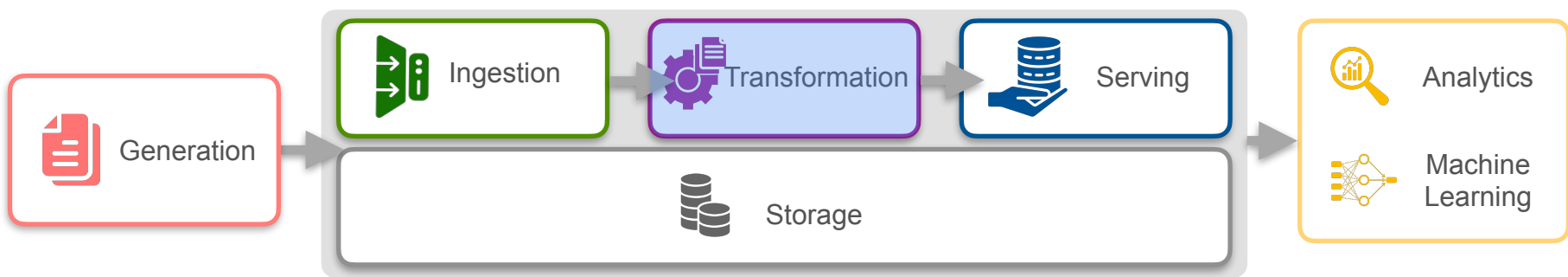


• Type of transformation

• Hardware specification

• Size of the data





Transformation

- Cleaning and combining data from multiple sources
- Convert data into a target schema based on a data model

- Apply transformations with SQL queries or Python
- Use a non-distributed or distributed processing tool



Analytics



Many fields	...	Many fields	...
...	...	...	...



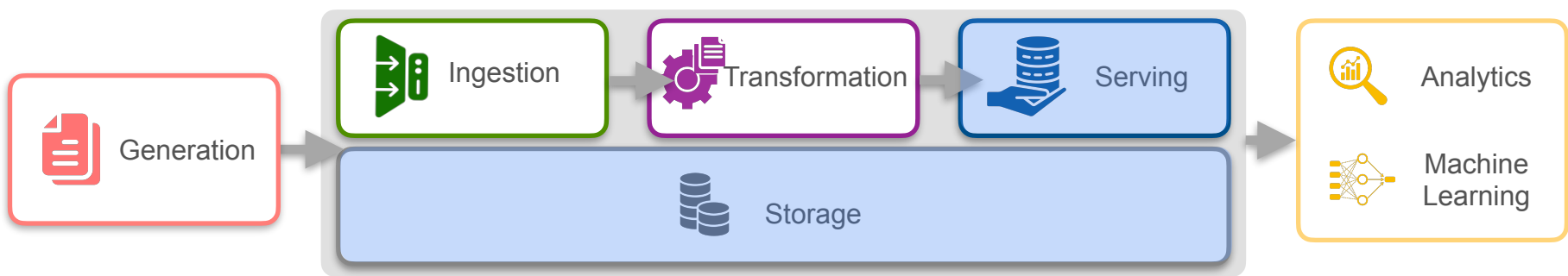
Process large amounts of data inside a cloud data warehouse to leverage MPP



Machine Learning

Transformations depend on end use case:

- Data exploration
- Training a machine learning model
- Making predictions



Data Warehouse

- Expects structured datasets with a well-defined schema
- Query optimizer returns results based on best execution plan
- Suitable for analytical workloads because it's based on columnar storage



Data Lake

- Built on top of low-cost object storage
- Supports ML & big data processing
- To prevent creating a data swamp, you can create a data catalog to track and manage data



Data Lakehouse



+



Low-cost & scalable storage

Structured query & data management

- Serves low-latency analytics and ML

Security



IAM



VPC



Network ACL



Route table

**Security
group**

Data Management

- Model your data



AWS Glue
Data Catalog

- Test & monitor
data quality



**great
expectations**



AWS Glue
Data Quality

DataOps

- Infrastructure
as code



HashiCorp
Terraform

Data Architecture

Orchestration



Apache
Airflow

Software Engineering



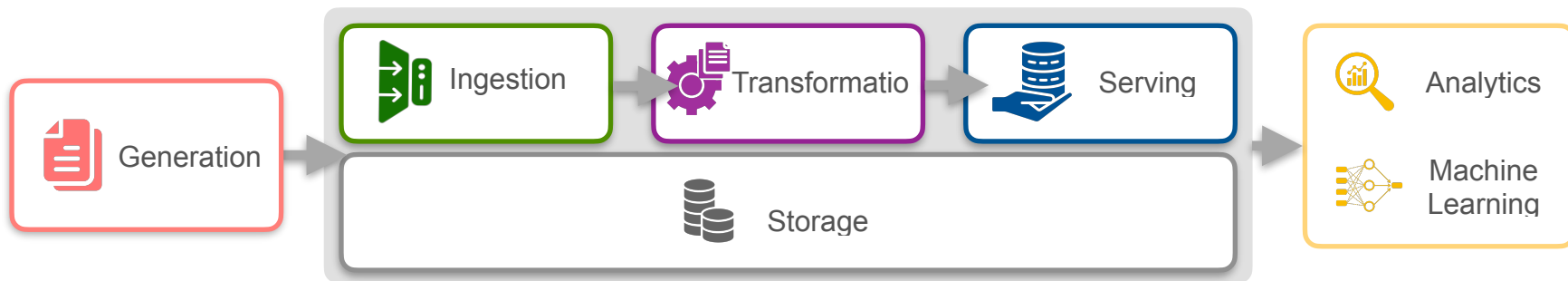
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Lab Walkthrough

Capstone Project (Part 1)

Capstone Lab

Solution to a business use case



Part 1 Create and configure the resources for the pipeline

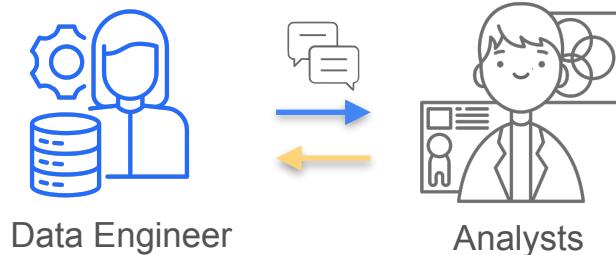
Part 2 Integrate data quality checks and orchestration,
& create data visualizations

Capstone Lab



- Offers subscription-based music streaming services
- Added a new feature allowing its clients to purchase and download music

GET	/users	Get Users
GET	/sessions	Get Sessions

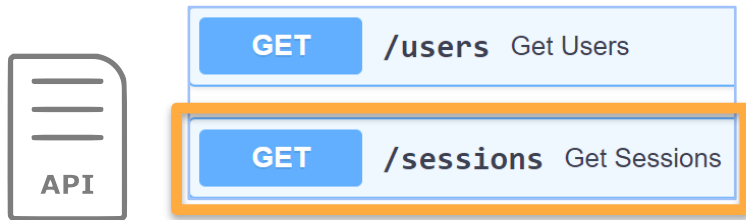


Building a data pipeline to serve the purchase data

Analyzing the purchase data

- with respect to the song and artist features
- how the purchases are evolving with time
- how the purchases are affected by the users' personal information

Data Sources



Response of “Get Users” Request

```
{
  "user_id": "a3141825-3a8c-4968-a3af-5362011ef7d5",
  "user_name": "Elizabeth",
  "user_lastname": "Carey",
  "user_location": [
    "46.32313",
    "-0.45877",
    "Niort",
    "FR",
    "Europe/Paris"
  ],
  "user_since": "2020-12-22T14:15:35.936090"
}
```

Response of “Get Sessions” Request

```
{
  "user_id": "6b287203-7cab-4f1a-b1a4-2b5076294682",
  "session_id": "04a5e8ac-1acd-48dc-88b9-651c4ddf489c",
  "session_items": [
    {
      "song_id": "TRXKAGX128F9342DD7",
      "song_name": "3 Cards",
      "artist_id": "AR475MP1187B9A5449",
      "artist_name": "The Balancing Act",
      "price": 1.03,
      "currency": "USD",
      "liked": true,
      "liked_since": "2023-01-27T08:29:54.970697"
    },
    {
      "song_id": "TRUKGBT128F4292C9B",
      "song_name": "Parisian Walls (gband Version_ Barcelona)",
      "artist_id": "ARP9HJX1187FB4E5DA",
      "artist_name": "Apostle Of Hustle",
      "price": 1.31,
      "currency": "USD",
      "liked": true,
      "liked_since": "2023-06-14T00:27:55.876873"
    }
  ]
}
```


Data Sources



GET	/users	Get Users
GET	/sessions	Get Sessions

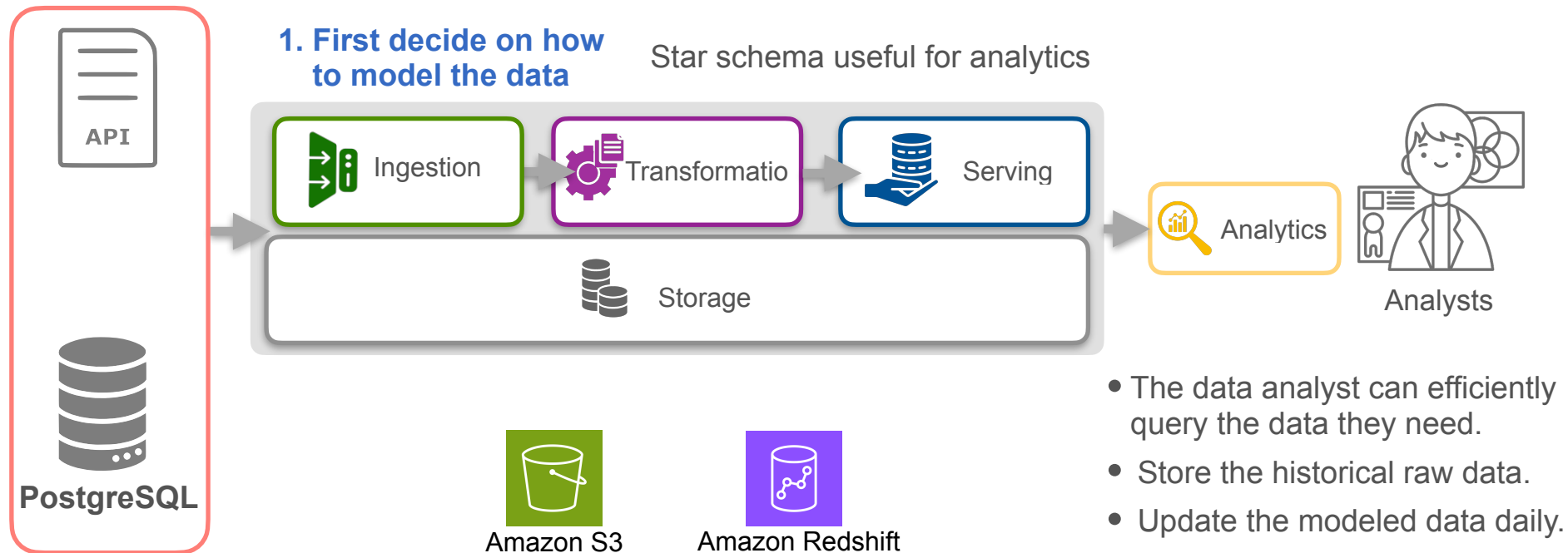


PostgreSQL

songs
track_id
title
song_id
release
artist_id
artist_mbid
artist_name
duration
artist_familiarity
artist_hottness
year
track_7digitalid

schema:
deftunes

Requirements



Data Model

Business process: Song purchases

Atomic Grain: Individual song item purchased within a session



API



PostgreSQL

1. First decide on how
to model the data

***Assuming all prices are in USD.
**If the prices were in different
currencies, convert all prices to
the same currency.*

dim_users
user_key (PK)
user_id
user_lastname
user_name
user_since
country_code

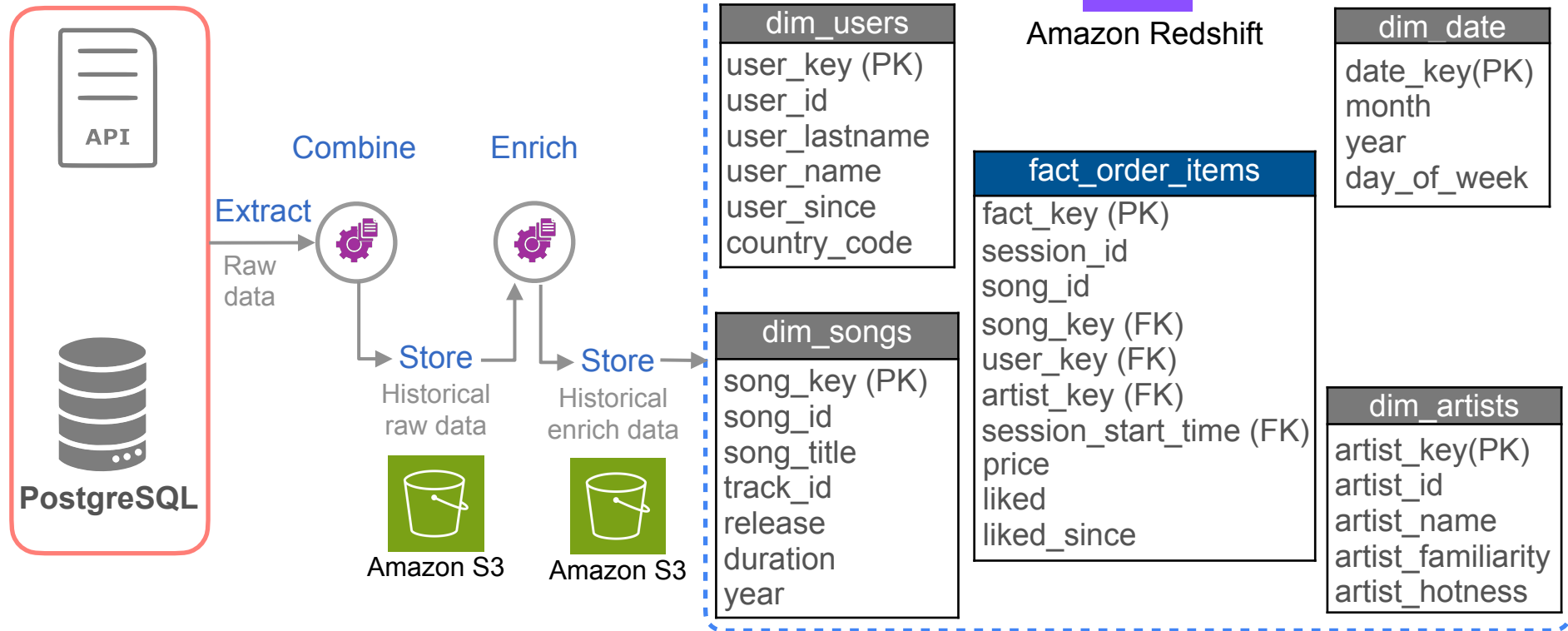
dim_songs
song_key (PK)
song_id
song_title
track_id
release
duration
year

fact_order_items
fact_key (PK)
session_id
song_id
song_key (FK)
user_key (FK)
artist_key (FK)
session_start_time (FK)
price
liked
liked_since

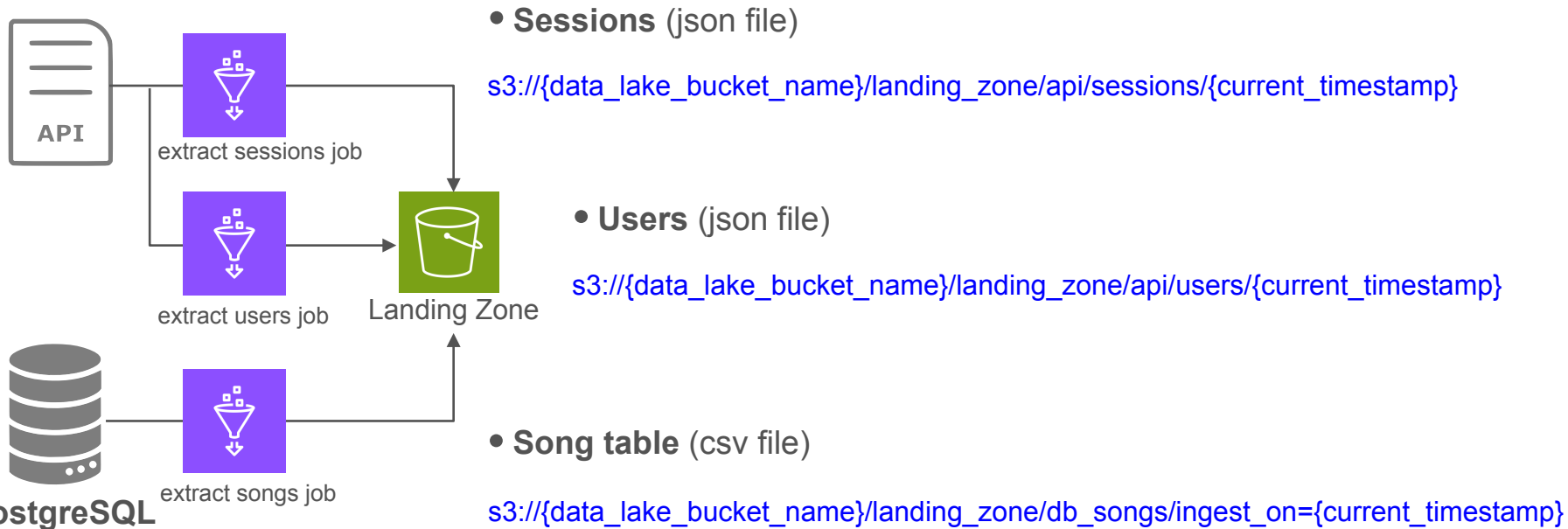
dim_date
date_key(PK)
month
year
day_of_week

dim_artists
artist_key(PK)
artist_id
artist_name
artist_familiarity
artist_hotness

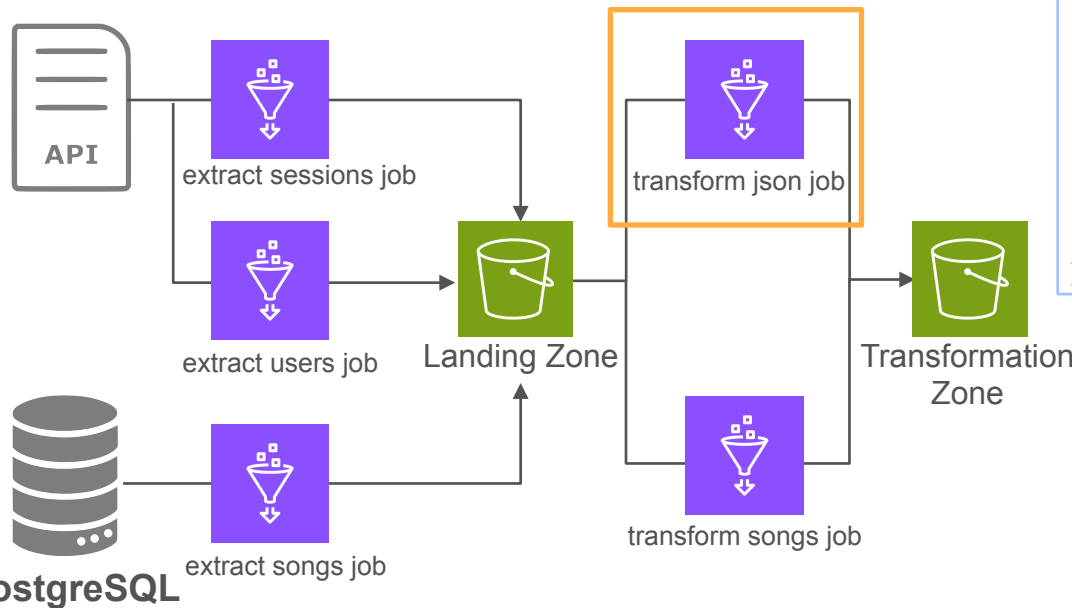
Data Model



Solution Architecture



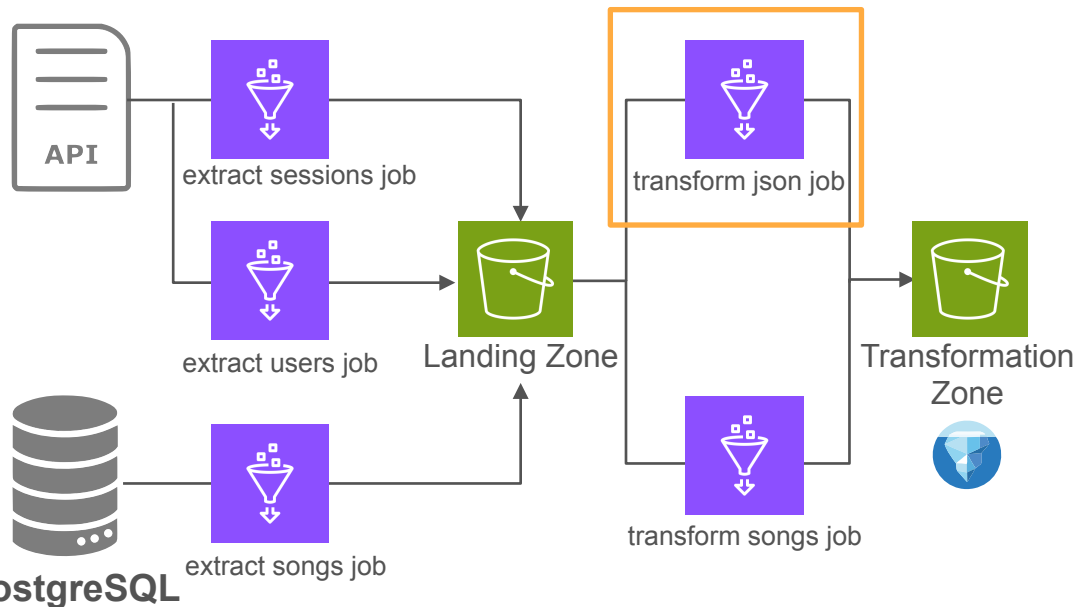
Solution Architecture



```
{
  "user_id": "a3141825-3a8c-4968-a3af-5362011ef7d5",
  "user_name": "Elizabeth",
  "user_lastname": "Carey",
  "user_location": [
    "46.32313",
    "-0.45877",
    "Niort",
    "FR",
    "Europe/Paris"
  ],
  "user_since": "2020-12-22T14:15:35.936090"
}
```

- ↓
1. Extract the users data into a DataFrame
 2. Unnest "user_location":
 - latitude
 - longitude
 - place name
 - country code
 - timezone
 3. Add a column indicating the ingestion time
 4. Add a column indicating the processing time

Solution Architecture

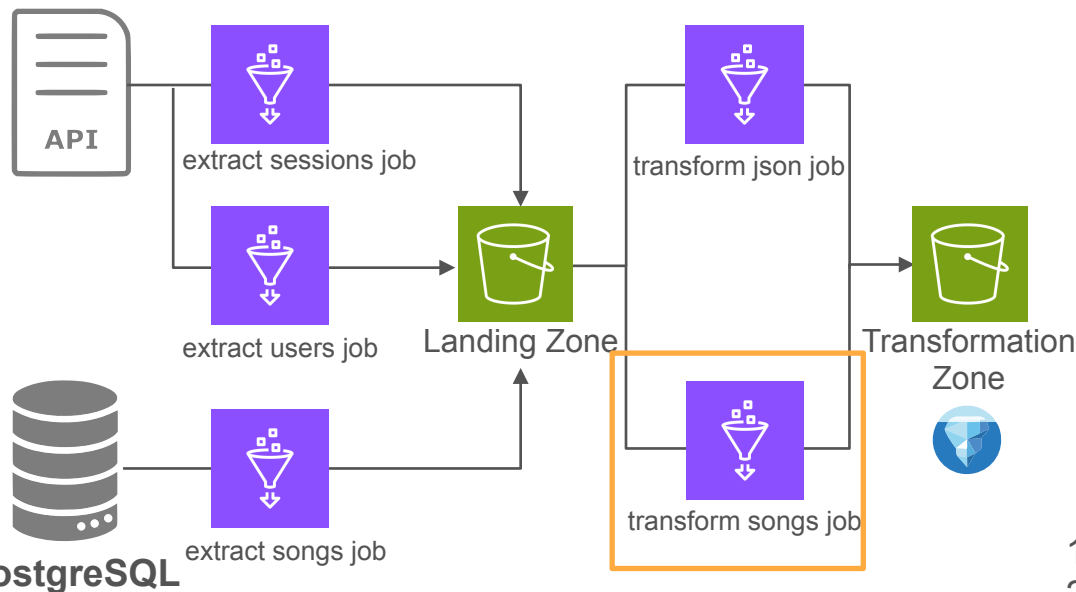


s3://{data_lake_bucket_name}/transform_zone/users
s3://{data_lake_bucket_name}/transform_zone/sessions

```
{  
  "user_id": "6b287203-7cab-4f1a-b1a4-2b5076294682",  
  "session_id": "04a5e8ac-1acd-48dc-88b9-651c4ddf489c",  
  "session_items": [  
    {  
      "song_id": "TRXKAGX128F9342DD7",  
      "song_name": "3 Cards",  
      "artist_id": "AR475MP1187B9A5449",  
      "artist_name": "The Balancing Act",  
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    },  
    {  
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      "artist_name": "Apostle Of Hustle",  
      "price": 1.31,  
      "currency": "USD",  
      "liked": true,  
      "liked_since": "2023-06-14T00:27:55.876873"  
    }  
  ],  
}
```

1. Extract the sessions data into a DataFrame
2. Unnest "session items"
3. Add a column indicating the ingestion time
4. Add a column indicating the processing time

Solution Architecture

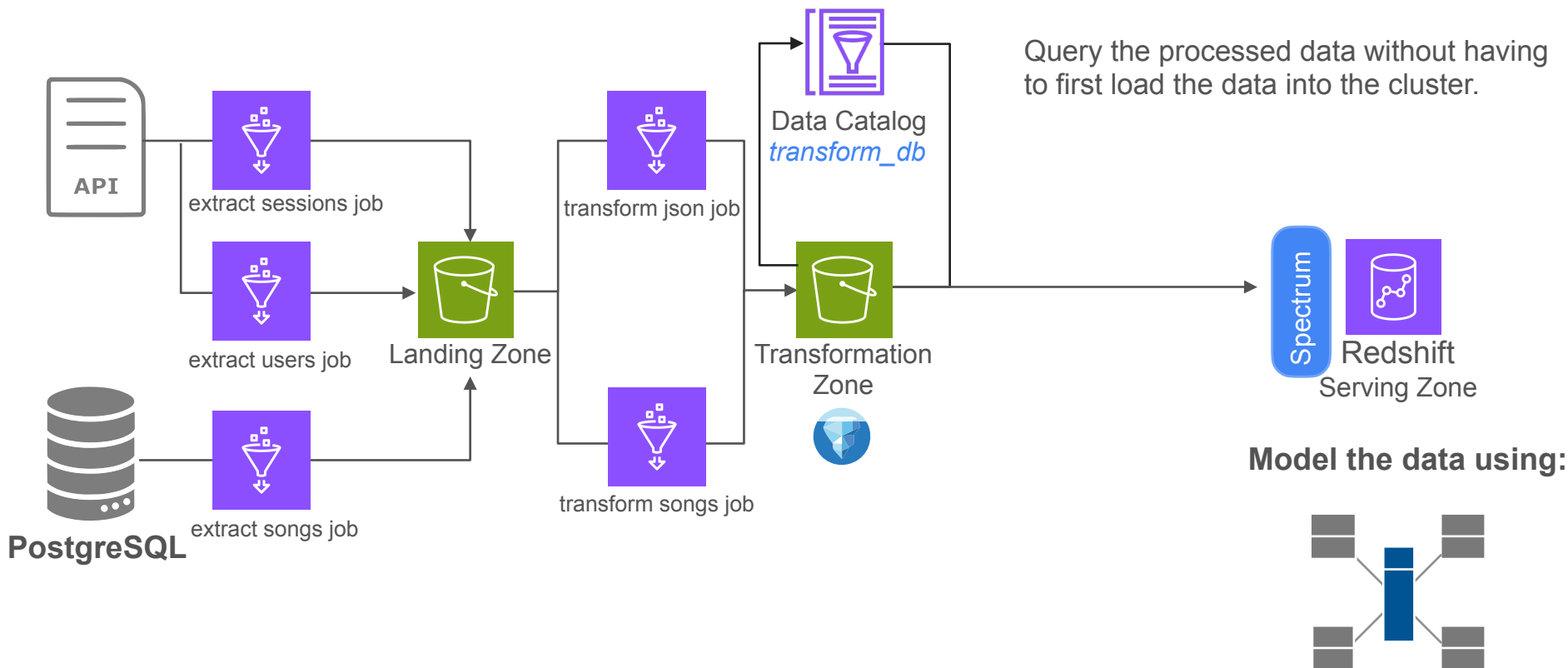


songs
track_id
title
song_id
release
artist_id
artist_mbid
artist_name
duration
artist_familiarity
artist_hottness
year
track_7digitalid

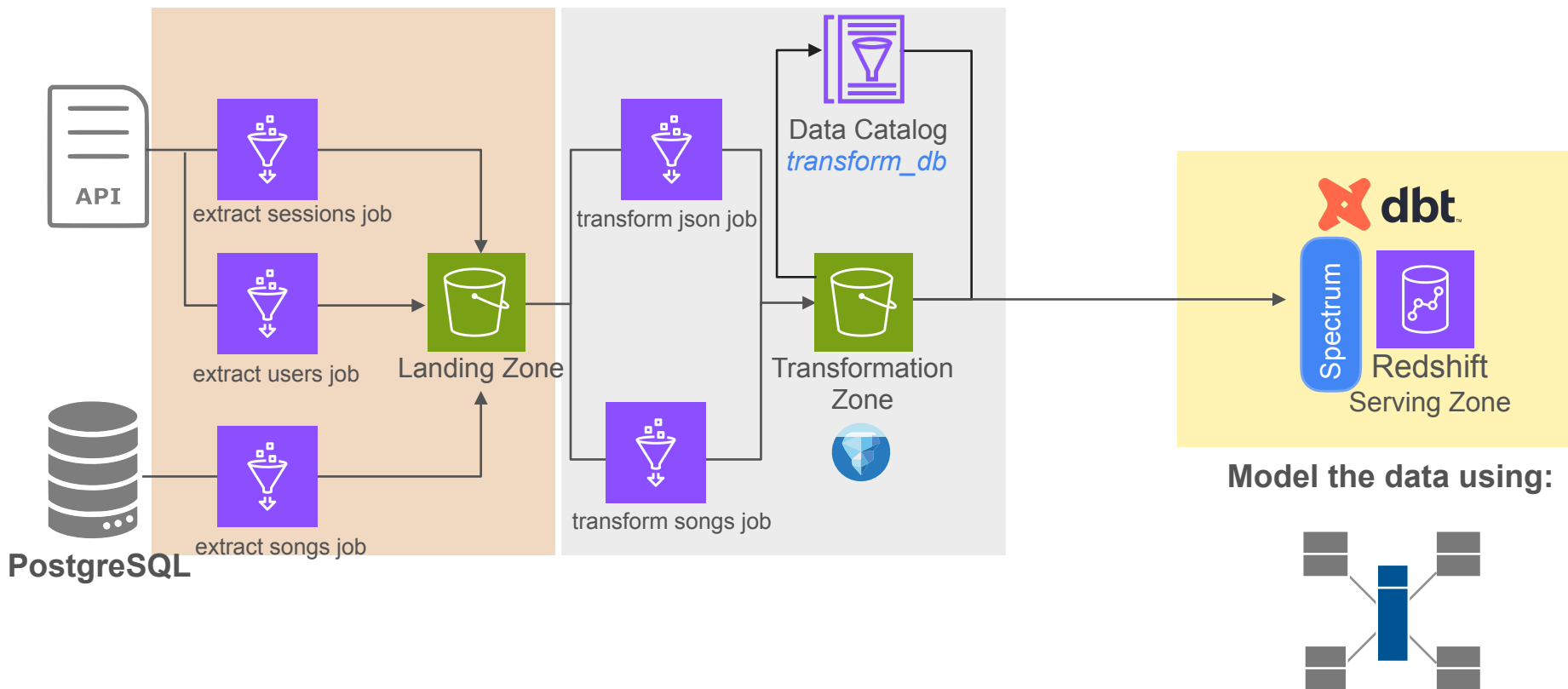


1. Extract the csv data into a DataFrame
2. Perform schema enforcement
3. Add a column indicating the ingestion time
4. Add a column indicating the name of the database source

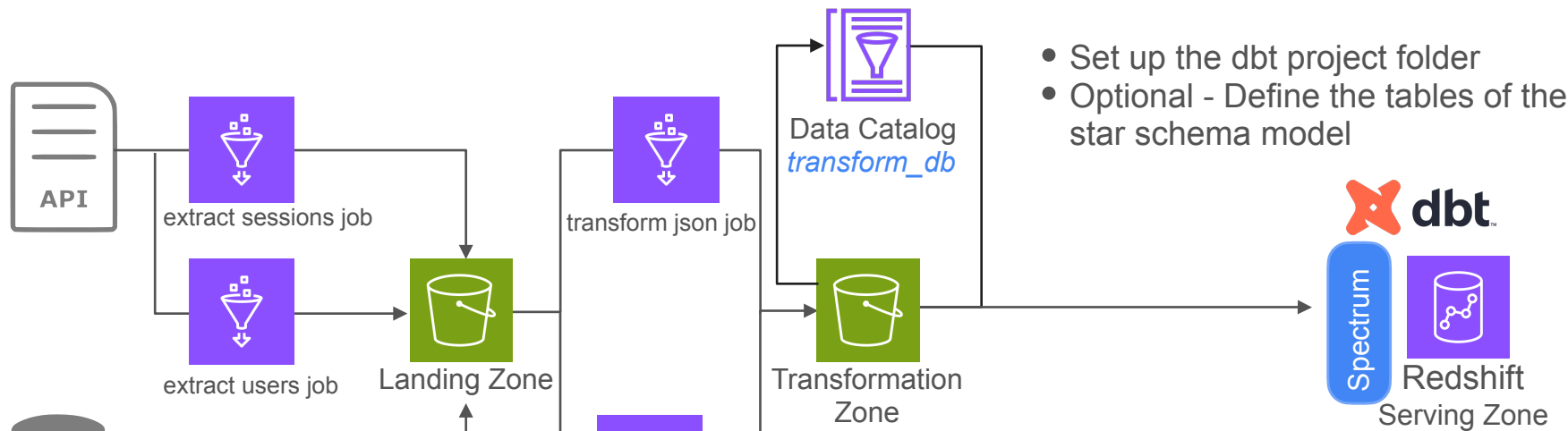
Solution Architecture



Solution Architecture



Capstone Lab - Part 1

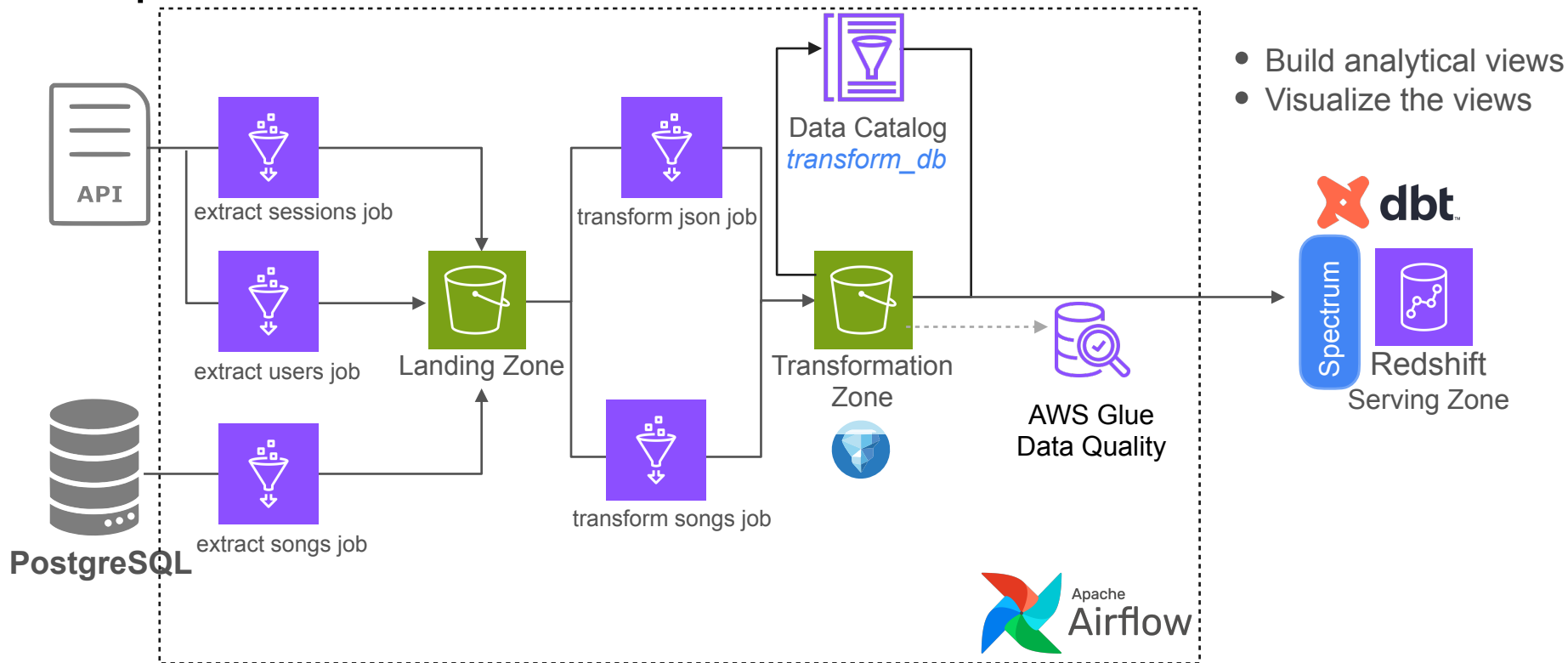


Model the data using:



Use Terraform to create the resources of the pipeline.

Capstone Lab - Part 2





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Lab Walkthrough

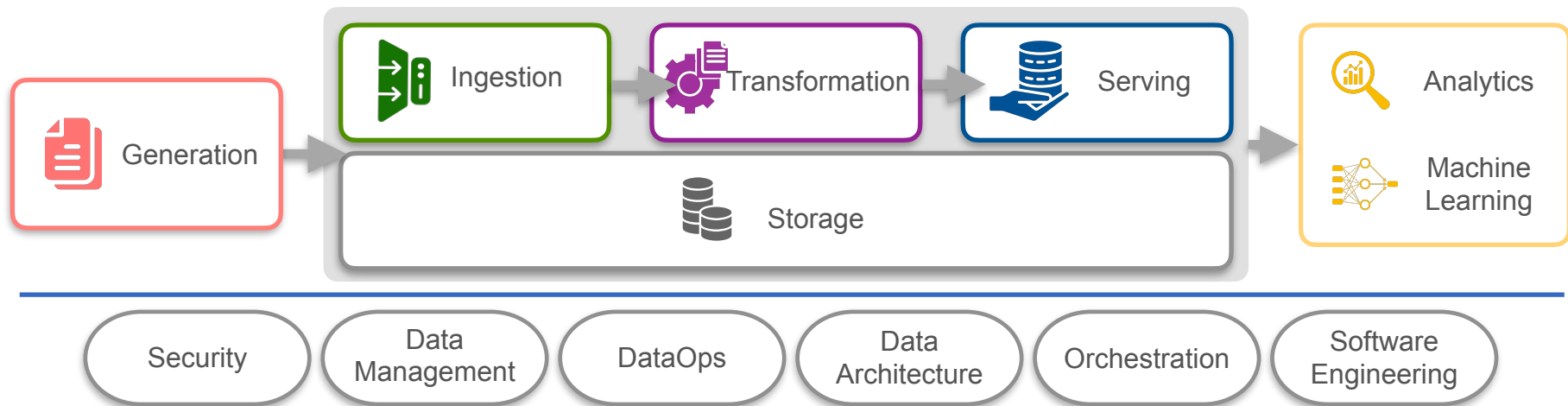
Capstone Project (Part 2)



DeepLearning.AI

Data Engineering

Conclusion & Thank You



The Role of the Data Engineer

Data Team



New role in the realm of
data and algorithms



Data Engineer



Software Engineer



Data Scientist



ML Engineer

The Role of the Data Engineer

Data Team



New role in the realm of
data and algorithms



Data Engineer



Software Engineer



Data Scientist ML Engineer

Operational discipline

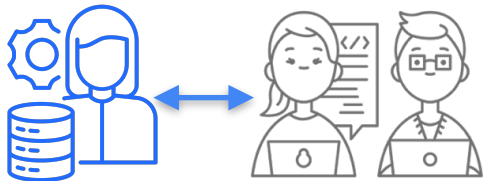
- Create or use systems that automate the development and operationalization of well-understood ML models
- Monitor data pipelines and quality

The Role of the Data Engineer

Data Team



New role in the realm of data and algorithms



Data Engineer Software Engineer

- Deep understanding in software and DE
- Expertise in streaming, data pipelines, data modeling, and data quality
- Boundaries between application backend systems and DE tools will be lowered



Data Scientist ML Engineer

Operational discipline

- Create or use systems that automate the development and operationalization of well-understood ML models
- Monitor data pipelines and quality